ISE 218: Fundamentals of Information Technology

Stony Brook Institute at Anhui University

Homework #2

Spring 2025

Assignment Due: March 07, 2025

Neatly write or type your answers to the following problems on a separate sheet of paper and submit a digital copy on Microsoft Teams by the due date.

For mathematical calculations you must show all work to receive full credit and simplify your answers as much as possible, giving a single number as your answer.

Please clearly print or type your **full name** and **ID #** on the first page of your submission.

For all questions involving sizes, you should use 2-based units of size. For example, 1 KB = $2^{10}$ bytes, not $10^3$ bytes.

1. [2 pts] Convert the binary number $1101101_2$ to decimal.

2. [2 pts] Convert the hexadecimal number $4F2A_{16}$ to binary.

3. [2 pts] Convert the binary number $1011010010011_2$ to hexadecimal.

4. [2 pts] Convert the binary number $110010.101_2$ to decimal.

5. [2 pts] Convert the decimal number $26_{10}$ to hexadecimal.

6. [2 pts] Convert the hexadecimal number $4FA_{16}$ to decimal.

7. [2 pts] Perform the following addition of 8-bit two's complement numbers: 11010110 + 11011111

8. [2 pts] What is the decimal value of the two's complement number 10001011?

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See next page for questions 9 to 13.

9. What decimal value does the 8-bit binary number 10000010 have if:

   a. [2 pts] it is interpreted as an unsigned number?
   b. [2 pts] it is on a computer using signed-magnitude representation?
   c. [2 pts] it is on a computer using two's complement representation?

10. Represent the decimal number −48 in binary using:

   a. [2 pts] 8-bit signed magnitude
   b. [2 pts] 8-bit two's complement

11. [2 pts] Convert the fixed-point binary value $1101011.1011_2$ to floating-point notation

12. [3 pts] Convert the IEEE 754 32-bit representation 10000000100100000000000000000000 into decimal.

13. [3 pts] Convert the number $-215.25_{10}$ into IEEE 754 32-bit notation.